

IIT Student Learning Outcome Assessment and Evaluation Application

Senior Capstone Project | Spring 2023

Project Overview

Client: Dr. Keiona Middleton, IIT Program Coordinator

Team: Avery Brockington, Brenden Wade, Samir Chowdhury, Dashaun Sorrells

Timeline: February - April 2023 (8 weeks effective development time)

My Role: Backend Database Developer, Integration Lead & De Facto Scrum Master

Special Circumstances: Mid-project pivot from original client/scope

Executive Summary

Note: This project began under extraordinary circumstances that tested our team's adaptability and resilience.

Our team's original capstone project—developing a sexual assault awareness and reporting application for Shaw Air Force Base—was cancelled at the start of Sprint 1 due to Department of Defense framework compliance requirements. While other capstone teams were beginning Sprint 2, we were reassigned to an entirely new project with a new client, new requirements, and an immediate two-week deficit in our timeline.

Despite starting from scratch mid-semester, our team successfully delivered a fully functional database system that met all client requirements within the compressed timeframe. This project showcases not only technical capabilities but also adaptability, project recovery skills, and the ability to deliver quality work under adverse conditions.

The University of South Carolina's IIT department needed a centralized system to manage student assessment data tied to program learning outcomes. Faculty were submitting assessment information through fragmented channels—emails, spreadsheets, and text files—making it nearly impossible to track trends, evaluate program effectiveness, or make data-driven curriculum decisions.

I led the development of a custom web-based database solution that consolidated assessment workflows into a single platform. The system features a MySQL relational database integrated

with a PHP-powered web interface, enabling faculty to submit, analyze, and visualize student performance data across multiple assessment types.

Key Impact:

- Eliminated manual data aggregation for 20+ faculty members
 - Standardized assessment submission across all IIT courses
 - Enabled real-time tracking of learning outcome achievement
 - Provided visual analytics for evidence-based curriculum improvements
 - Delivered complete, functional system despite 2-week development deficit from mid-project restart
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The Challenge

Problem Statement

The IIT program lacked any standardized method for collecting, storing, or analyzing learning outcome assessment data. Faculty submitted assessment results through various disconnected channels, creating several critical problems:

- **Data Fragmentation:** Assessment information scattered across emails, Excel files, and text documents
- **No Historical Tracking:** Unable to compare student performance across semesters or academic years
- **Time-Intensive Analysis:** Faculty spent hours manually compiling data for program reviews
- **Limited Visibility:** No centralized view of which learning outcomes students were meeting or missing
- **Compliance Risk:** Difficulty demonstrating assessment practices during accreditation reviews

Project Constraints

- **Compressed Timeline:** Effective 8-week development window after mid-project restart (vs. standard 10 weeks for other teams)
- **Late Start Disadvantage:** Began Sprint 1 when peer teams were completing Sprint 2
- **Budget:** No funding for commercial solutions (required custom development)
- **Scope:** Single-department implementation (not university-wide)
- **Users:** Faculty with varying technical proficiency
- **Project Uncertainty:** Complete requirements reset after initial client cancellation

Success Criteria

The client defined success as a system where faculty could:

1. Submit assessment data through an intuitive web interface
 2. Generate reports filtered by date range, course, and learning outcome
 3. Visualize trends to identify areas where students struggle
 4. Access the system securely with role-based permissions
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Solution & Technical Implementation

System Architecture

I designed and implemented a three-tier web application:

Backend Database (MySQL)

- Designed normalized relational database with 8 core tables
- Created entity-relationship model to eliminate data redundancy
- Implemented referential integrity constraints
- Optimized queries for report generation performance

Integration Layer (PHP)

- Developed server-side processing for form submissions
- Built secure data validation and sanitization routines
- Created dynamic SQL query generation for custom reports
- Implemented session management for user authentication

Frontend Interface (HTML/CSS)

- Designed responsive web forms for assessment data entry
- Created interactive dashboards for data visualization
- Built custom reporting interface with flexible filtering
- Ensured accessibility standards compliance

Hosting & Deployment

- Deployed on 000webhost cloud platform
- Configured secure database connections
- Implemented backup procedures

Technical Stack

Backend: MySQL 5.7

Integration: PHP 7.4

Frontend: HTML5, CSS3

Hosting: 000webhost

Tools: Lucidchart (ERD modeling), WordPress (prototyping)

Key Features Delivered

1. Intelligent Data Submission Workflow

- Faculty log in with secure credentials
- Guided prompts collect structured assessment data:
 - Course information and semester
 - Learning outcome being assessed
 - Assignment/assessment type used
 - Success criteria (e.g., "70% of students earn C or above")
 - Actual student performance results
- Automatic success/failure calculation based on criteria
- Freeform text field for contextual analysis

2. Dynamic Reporting Engine

- Filter reports by date range (semester or academic year)
- Select specific learning outcomes to analyze
- View aggregate performance across multiple courses
- Export data for external analysis

3. Visual Analytics Dashboard

- Charts showing learning outcome achievement rates over time
- Course-level performance comparisons
- Identification of consistently challenging outcomes
- Trend analysis for curriculum improvement planning

4. Data Management

- Edit previously submitted assessments
- Maintain historical records for accreditation
- Support for multiple assessment types (tests, rubrics, surveys, projects)

Development Process

Project Pivot & Recovery

Original Project (Weeks 1-3): Our team was initially assigned to develop a sexual assault awareness and reporting mobile application for Shaw Air Force Base. The application aimed to provide military personnel with confidential resources, reporting mechanisms, and support services in a secure, stigma-free environment.

Mid-Project Cancellation: At the beginning of Sprint 1, Shaw Air Force Base withdrew from the project due to Department of Defense security framework and compliance requirements that couldn't be met within an academic setting. This left our team without a client, requirements, or deliverables while other capstone teams were advancing into Sprint 2.

Rapid Pivot: Within one week, we:

- Connected with Dr. Keiona Middleton and learned the IIT department's assessment challenges
- Conducted stakeholder interviews to understand requirements
- Drafted a new project charter and scope
- Reorganized our sprint schedule to compress 10 weeks of work into 8 weeks
- Aligned our team on new technical stack and deliverables

This experience in project recovery and rapid requirements gathering became one of the most valuable aspects of the capstone experience, demonstrating real-world agility beyond typical academic projects.

Agile Methodology Under Pressure

The project followed an accelerated four-sprint Agile development cycle with weekly client check-ins and daily standups. While the scrum master role was intended to rotate every two weeks among team members, I took ownership of consistently producing all weekly team status reports, coordinating sprint planning, and maintaining project documentation throughout the entire project. This additional leadership responsibility beyond my technical duties ensured continuity and kept the team aligned despite the compressed timeline and competing academic obligations.

Our team worked with exceptional focus and coordination, often dedicating 10-15 hours per week (well above the standard 5-7 hours for other teams) to close the two-week gap created by the mid-semester restart.

Sprint 1: Accelerated Discovery & Design (1.5 weeks)

- Conducted rapid stakeholder interviews to gather requirements
- Created Entity-Relationship Diagram (ERD) for database structure
- Designed wireframe mockups of user interface
- Received client approval on data model and UX flow
- **Compressed typical 2-week sprint into 1.5 weeks to recover timeline**

A wireframe mockup of a web page titled "USC IT Assessment & Evaluation". The page is displayed within a browser window with a title bar "A Web Page" and a URL bar showing "https://". The page layout includes a red header bar with a placeholder icon (a square with an 'X') on the left and the title "USC IT Assessment & Evaluation" on the right. Below the header, the text "USC Authentication Service" is visible. The main content area contains a "USC Staff Login:" section with two input fields: "Username" and "Password". The page is framed by a grey border, and a small double-slash icon is visible in the bottom right corner of the frame.

A Web Page

https://

Please select an academic year and class:

2022-2023

ITEC 564

A Web Page

https://

Students

Grades

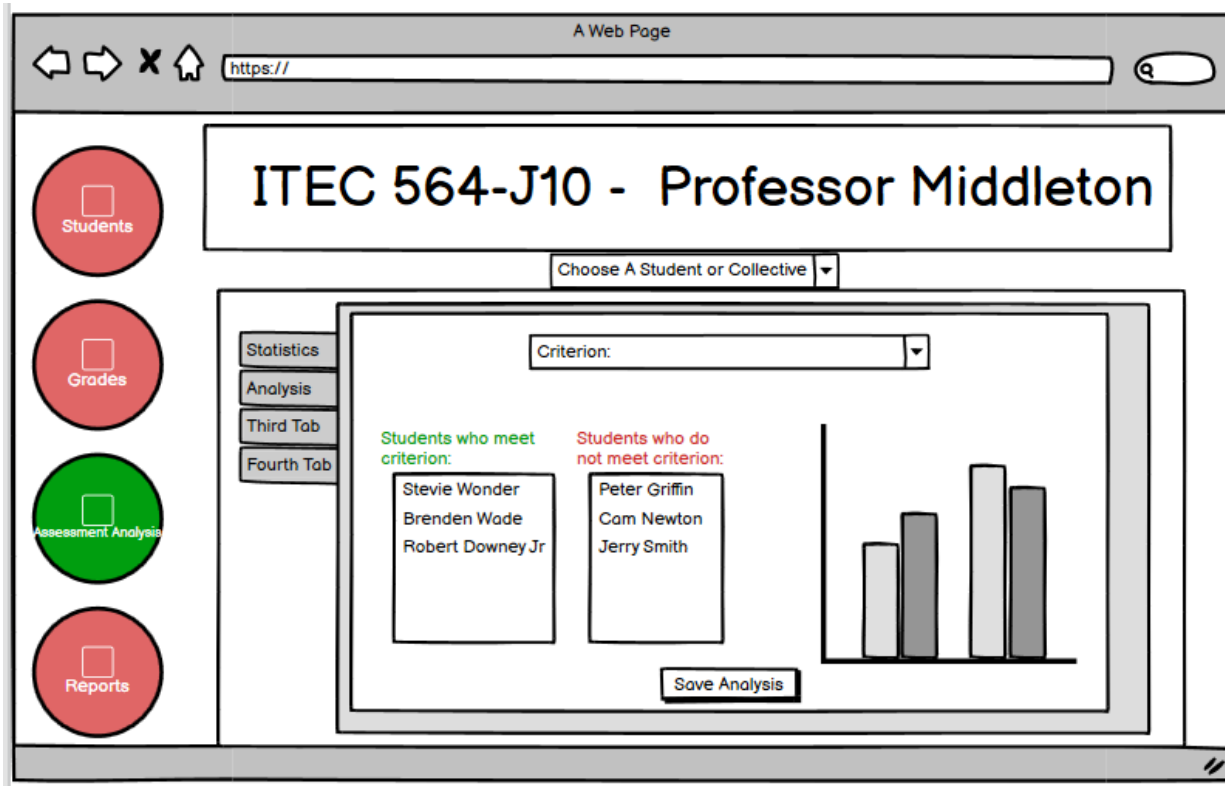
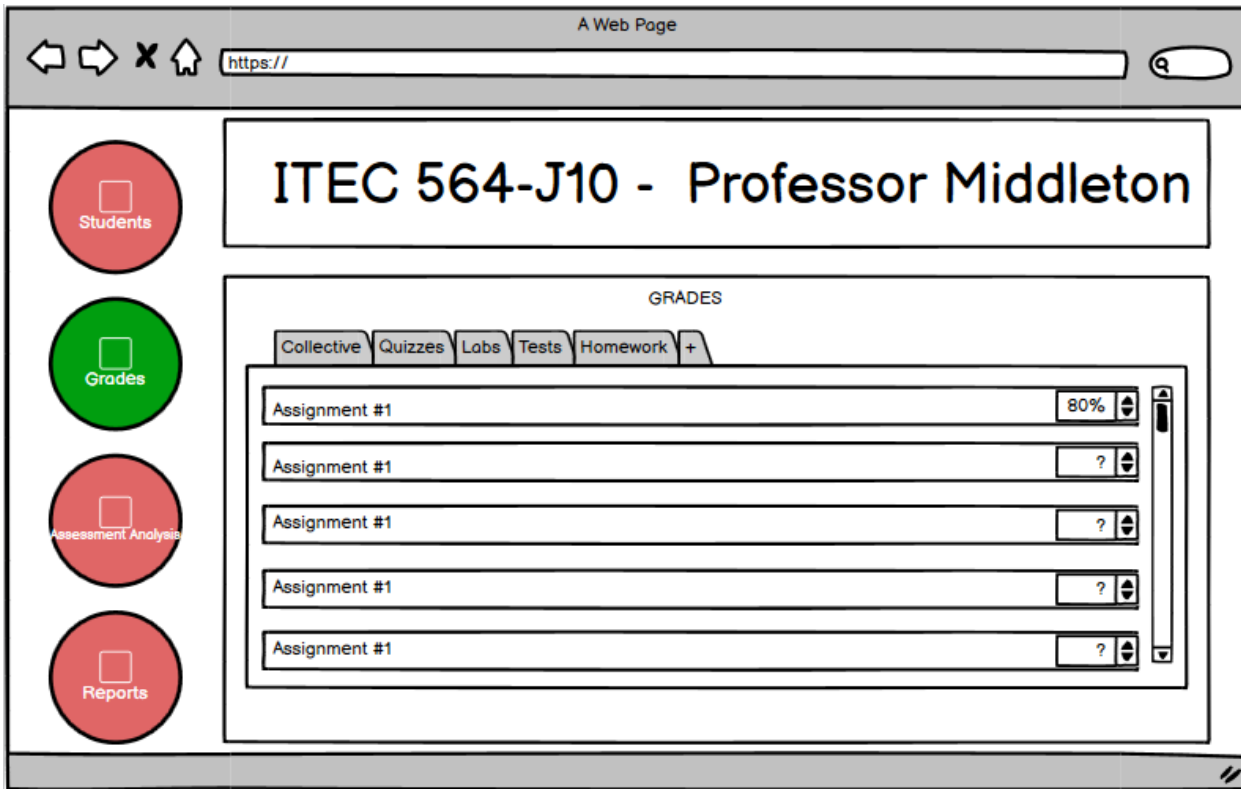
Assessment Analysis

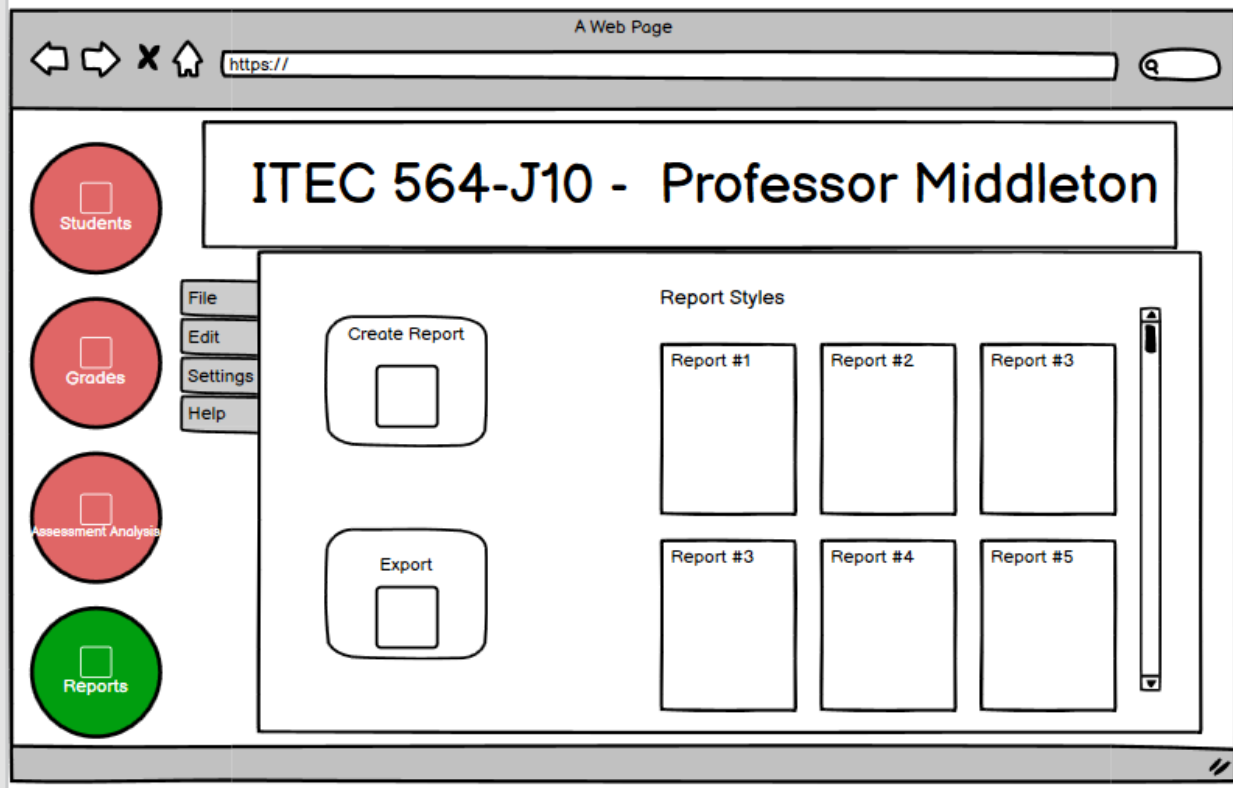
Reports

ITEC 564-J10 - Professor Middleton

Last activity...

Created an Assessment Report on Assignment #3	February 12, 2023





Sprint 2: Foundation Development (2 weeks)

- Built MySQL database with normalized table structure
- Developed initial HTML forms for data entry
- Created low-fidelity prototype demonstrating core workflow
- Researched PHP integration approaches

Sprint 3: Integration & Refinement (2 weeks)

- Connected frontend forms to backend database via PHP
- Implemented data validation and error handling
- Developed reporting functionality
- Conducted usability testing with faculty

Sprint 4: Finalization & Deployment (2 weeks)

- Resolved database connectivity issues
- Migrated to 000webhost for stable hosting
- Refined UI styling and user experience
- Delivered final product and documentation

Challenges & Solutions

Challenge 0: Complete Project Reset Mid-Semester Our original Shaw Air Force Base project was cancelled due to DoD compliance requirements, leaving us two weeks behind all other capstone teams with no client, no requirements, and no deliverables.

Solution: Led rapid requirements gathering with new client within 5 days. Reorganized team responsibilities and accelerated sprint timeline. Maintained team morale by focusing on what we could control: work ethic, communication, and technical quality. This adversity became an opportunity to demonstrate adaptability—a skill that's often more valuable than perfect execution in ideal conditions.

Challenge 1: Database-Frontend Integration Initial attempts to connect the PHP forms to the MySQL database failed due to server configuration issues on our development environment.

Solution: Migrated the entire application to 000webhost, which provided a properly configured LAMP stack. This resolved connectivity issues and allowed seamless data flow between frontend and backend.

Challenge 2: Compressed Timeline with Quality Expectations Starting Sprint 1 when other teams were finishing Sprint 2 meant we had 20% less development time while maintaining the same quality standards and deliverable expectations.

Solution: Prioritized ruthlessly based on client's critical success factors. Identified which features were essential for MVP vs. nice-to-have. Increased weekly time commitment to 10-15 hours (vs. standard 5-7). Leveraged our team's diverse skills efficiently, I focused exclusively on backend and integration while frontend developers specialized in UI. No wasted effort on scope creep or over-engineering.

Challenge 3: Complex Data Relationships Modeling the relationships between courses, learning outcomes, assessments, and results required multiple ERD iterations to eliminate redundancy while maintaining flexibility.

Solution: Collaborated closely with the client to understand assessment workflows. Simplified the initial design by removing two redundant tables, which improved both data entry efficiency and query performance.

Challenge 4: Report Flexibility Requirements The client needed to generate reports with various combinations of filters (date ranges, specific outcomes, multiple courses) without creating dozens of separate report templates.

Solution: Implemented dynamic SQL query generation in PHP that constructs queries based on user-selected filters, providing infinite report customization possibilities through a single interface.

Results & Impact

Delivered Product

The final database system successfully met all client requirements and has been deployed for faculty use. Key capabilities include:

- **Secure web-based interface for assessment data submission**
- **Automated success criteria evaluation**
- **Flexible reporting with date range and field selection**
- **Visual representation of learning outcome trends**
- **Historical data preservation for accreditation**
- **Delivered on time despite starting two weeks behind schedule.**
- **System architecture adopted as foundation for continued development in subsequent semesters for the IIT Capstone Course.**

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Professor Login

Username:

Password:

Login →

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Please select an academic year and class

Academic Year:

2022-2023 ▾

Class:

Class 1 ▾

Let's Go →

Assessment and Evaluation

Home

Grades

Analysis

View

Last Activity

Insert your user's last activity here.

Assessment and Evaluation

Home

Grades

Analysis

View

Grade Analysis Report

Please select your requirements below, and click calculate to generate a report.

Assess the advantages and disadvantages of managing information technology projects in the organizational context. ▾

Assess the advantages and disadvantages of managing information technology projects in the organizational context.

Evaluate and apply current technologies and technical concepts, practices, and skills from core IT courses to satisfy the needs of the clients and users in the completion of a successful IT project.

Initiate, plan, execute, monitor and control, and close a successful project from the start through completion.

Successfully manage project processes.

Successfully develop the project product outcomes.

Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.

Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.

Communicate effectively in a variety of professional contexts.

Results

Powered by 000webhost

Results Description:

Results:

SUBMIT

Assignment	Description	Grade
	test	100
Name	this is a test again	100
Name	this is a test again	100
Name	test	100
Name	test	100
Name	test	100
Name	test	100
Name	test	100
Name	test	100
Name here	test	55

Powered by 000webhost

Assessment and Evaluation

Home

Grades

Analysis

View

Student Grades

Upload

Faculty:

Course:

Online/F2F:

Learning Outcome:

Success Criteria:

Assignment:

Description:

Results Description:

Results:

SUBMIT

Assessment and Evaluation

[Home](#)[Grades](#)[Analysis](#)[View](#)

Grade Analysis Report

Please select your requirements below, and click calculate to generate a report.

Assess the advantages and disadvantages of managing information technology projects in the organizational context.

Criteria:

Check

The criteria was met by 88.88888888889% of students.

Results

Client Feedback

Dr. Middleton expressed satisfaction with the system's ability to streamline previous manual processes. The standardized submission workflow ensures consistency across all faculty members, and the reporting features provide unprecedented visibility into program-wide student performance.

Long-term Impact: The database framework was deployed for active faculty use and subsequently served as the foundation for continued development by future capstone cohorts. The IIT department adopted our system architecture as the baseline for ongoing enhancements, demonstrating that our compressed-timeline solution was robust enough to support iterative expansion rather than requiring a complete rebuild.

Personal Contribution

As the backend developer, integration lead, and de facto scrum master, I was responsible for:

- **Complete database design and implementation**
- **All PHP integration code**
- **SQL query optimization**
- **Deployment configuration and troubleshooting**
- **Technical documentation**
- **Weekly status reports and sprint coordination for the entire project duration**
- **Facilitating team standups and maintaining project tracking tools**
- **Primary liaison between development team and client**

This dual role of technical lead and project coordinator significantly strengthened both my full-stack development skills and project management capabilities, particularly in database design, server-side scripting, team leadership, and translating business requirements ranged from clear to ambiguous into technical solutions.

Lessons Learned & Professional Growth

Technical Insights

Project Recovery is a Critical Skill: The mid-project cancellation could have derailed our entire semester. Instead, it taught me that adaptability and rapid requirements gathering are as important as technical skills. Within one week, we pivoted from a mobile security application to a web-based database system—completely different tech stacks, use cases, and clients. This

experience mirrors real-world scenarios where projects get cancelled, requirements change dramatically, or timelines compress unexpectedly.

Database Design Complexity: Effective database normalization requires deep understanding of the business domain. Our initial ERD was over-engineered; simplifying it improved both usability and performance. The lesson: prioritize practical functionality over theoretical perfection.

Integration Challenges: Connecting disparate technologies (HTML, PHP, MySQL) taught me the importance of environment configuration and debugging systematic connection issues. Migrating to a properly configured hosting platform was crucial to success.

Security Considerations: Even in an internal university tool, implementing proper authentication, data validation, and SQL injection prevention is critical. I gained practical experience with secure coding practices that will inform all future projects.

Agile Methodology Benefits

Working in sprints with regular client feedback proved invaluable. Early course corrections based on Sprint 1 feedback prevented major rework later. Taking on the scrum master responsibilities for the full project duration—beyond the intended two-week rotations—taught me the importance of consistent leadership and documentation in keeping distributed teams aligned. The weekly standup meetings and status reports I facilitated kept the team synchronized despite competing academic obligations and varying schedules.

Soft Skills Development

Client Communication: Learning to translate technical concepts for non-technical stakeholders improved my ability to explain complex systems in accessible language.

Team Coordination: Balancing development work across four team members required clear task delegation and consistent communication about integration points. Serving as the continuous scrum master taught me how to facilitate productive meetings, track blockers, and maintain momentum even when team members faced competing priorities.

Adaptability: When our original project fell through mid-semester, pivoting quickly to a new client with entirely different technical requirements demonstrated resilience and project recovery skills. Rather than viewing this as a setback, I approached it as an opportunity to prove we could deliver under pressure with limited runway.

Future Enhancements

Given additional time and resources, the system could be enhanced with:

Enhanced Security

- Multi-factor authentication for sensitive data access
- Role-based permissions (admin vs. faculty vs. read-only)
- Comprehensive audit logging of data modifications
- Encrypted data transmission (HTTPS implementation)

Expanded Analytics

- Predictive modeling to identify at-risk learning outcomes.
- Comparative benchmarking across similar programs
- Automated email alerts when success criteria are not met.
- Integration with university student information systems

User Experience Improvements

- Mobile-responsive design for tablet access
- Batch data import from existing spreadsheets.
- Customizable dashboard widgets
- Export capabilities (PDF, CSV, Excel)

Scalability Considerations

- Migration to enterprise database platform
- Caching layer for improved performance
- Microservices architecture for university-wide deployment

Note: Following our initial deployment, subsequent capstone cohorts built upon our database framework, validating that our architectural decisions were sound and extensible. The system continues to evolve based on the foundation we established under the compressed timeline.

Project Artifacts

Note: Per university policy, all project deliverables and source code are proprietary to the University of South Carolina and cannot be publicly shared. The following documentation was produced during development:

Documentation

- Complete Project Charter with scope and requirements
- Entity-Relationship Diagram (ERD) showing database structure
- User Interface wireframes and high-fidelity mockups
- Sprint retrospectives and burndown charts
- Technical documentation for future maintenance

Presentations

- Capstone Expo Poster (April 2023)
- Client demonstration and handoff materials
- Final presentation to faculty stakeholders

I can discuss technical implementation details, architectural decisions, and lessons learned in interviews, though I cannot provide access to source code or detailed system documentation due to institutional ownership.

Reflection

This capstone project provided invaluable experience in full-stack web development, database design, and client-focused software engineering. Beyond technical skills, I developed stronger project management capabilities, learned to navigate team dynamics under deadline pressure, and gained confidence in delivering functional software that solves real organizational problems.

The most significant lesson came from adversity itself. When our Shaw Air Force Base project was cancelled and we started Sprint 1 while other teams were finishing Sprint 2, we had every reason to request timeline extensions or reduced scope. Instead, our team chose to view the two-week deficit as a challenge to overcome through superior work ethic, efficient coordination, and technical focus. We delivered a complete, functional system that met all client requirements on the original deadline.

The most rewarding aspect was seeing faculty immediately recognize the value of our solution despite the compressed development timeline. Knowing that this system continues to serve the IIT department—and that our architecture was robust enough to become the foundation for future enhancements by subsequent student cohorts—validates that we delivered production-quality work under extraordinary circumstances.

This project taught me that constraints often reveal capability. Given perfect conditions, any team can deliver. But delivering quality work when dealt with an objectively unfair situation? That is where real professional growth happens.

For more information about this project or to discuss technical implementation, please contact me.